



Working with Medical Volumes

Often, you need to do more than just view your data. For example, you may need to extract specific slices for future processing.

Navigate to the Module 2 folder and open the **WorkingWithMedicalVolumes.mlx** file to import a medical volume, view the metadata, and extract slices from the volume.

We recommend working with the code in MATLAB to gain experience and experiment and practice. A static version of the code is provided below. The static version does not include the interactive tools needed to run the code.

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- `VoxelSpacing` and `SpatialUnits` are used to convert voxels to real-world measurements.
- `Planemapping` tells you how 3D voxel dimensions are mapped to the patient coordinate system.
- `VolumeGeometry` contains information needed to orient the image to real-world coordinates.
- `Modality` is how the image was acquired, in this case, MR or Magnetic Resonance.

View the Data

Use `orthosliceviewer` to quickly display volumetric image data. A new window for viewing and navigating through the image will open. Meta-data in the `medicalVolume` is used to display the image in the proper orientation and scale.

- Set the `displayrange` property to change the default display limits.
- Manually adjust the window level and window contrast by left-clicking and dragging the mouse on an image.
- Click and drag the crosshairs to move through the slices as you did in the Medical Image Labeler app.

By default, the window level and center from the meta-data are used for the display range, but you can adjust this if needed.

```
mindisplay = min(mri.Voxels, [], "all");
maxdisplay = max(mri.Voxels, [], "all");
orthosliceviewer(mri, displayrange = [mindisplay, maxdisplay]);
```

Extract 2D Images

After viewing your volume, you'll likely find several slices of interest that you need to analyze. The `extractSlice` function extracts a 2D image from the volume. Provide the volume, slice number and plane to extract as inputs. The function also returns the real-world position of the top-left pixel of the extracted slice and the spacing between voxels. When spacing in the X and Y directions is not equal, use the spacing information to display the slice with the correct scale

Use the controls below to extract a slice and view the result. You'll get an error if `sliceNum` exceeds the number of slices in that plane. If that happens, just decrease the value.

```
plane = "Sagittal";
sliceNum = s0;
[img, position, spacing] = extractSlice(mri, sliceNum, plane);

% Check if the spacing is equal
spacing
imshow(img, [])
```

If the spacing in X and Y is not the same, the image may appear distorted. To properly scale the image, specify the real-world image extents in X and Y.

```
xMax = size(img, 2)*spacing(2);
yMax = size(img, 1)*spacing(1);
```